Lack of Island sensitivity transfers: Vietnamese L1 effects on L2 English sentence processing

Linh Pham, Zuzanna Fuchs, Elsi Kaiser- University of Southern California- llpham@usc.edu

Background: Research shows that bilinguals' linguistic systems are not separate but interconnected (e.g. [5]), interacting with each other during language processing. Empirical studies provide support for this interaction at different levels of linguistic representation, including phonological (e.g.[3]) and lexical levels (e.g. [1]). However, evidence is mixed for interaction at the structural level, especially in the case of abstract constraints that are not reflected in surface word order.

This study contributes to ongoing debates on language transfer in real-time processing by investigating whether the absence of the relative clause (RC) island constraint – an abstract feature not reflected in surface word order – in a speaker's L1 affects the processing of island structures in L2, while controlling for language immersion in the L2 to maximize potential L1 influence.

Prior work on L1 processing shows that *island constraints* in processing block the **A**ctive gap-**F**illing **S**trategy (AFS) [6], which normally favors resolving dependencies early (e.g. placing the gap at position (a) in ex.(1) instead of waiting for the actual gap at position (b)), by preventing parsers from positing gaps within certain syntactic boundaries (e.g. inside RC islands, ex.2). However, some languages like Vietnamese allow violations of this constraint (ex.3) [8]. Crucially, Vietnamese (SVO, head-initial RCs) has the same surface word order as English but, unlike English, lacks RC island constraints. This contrast allows us to test whether the absence of an abstract constraint in the L1 affects L2 processing of that structure, while controlling for the effects of word order.

Experiments: Following [7], we ran self-paced reading studies using a 2×2 design crossing island status (island vs. non-island) and plausibility (plausible vs. implausible) (see Table 1,2 for Vietnamese, English stimuli). If the parser is sensitive to the RC island constraint, it should not posit a gap in the island conditions and thus should show no plausibility effects in island conditions – in contrast to non-island conditions, which are predicted to show effects of plausibility.

Exp.1 (n = 60, Vietnamese native speakers) tests whether L1 processing of Vietnamese shows signs of RC island sensitivity (Fig.1). Using self-paced reading, we find slower RTs in implausible than plausible conditions at both spillover regions after the critical word (e.g. *viết* 'write') ($p<.0001$). Crucially, this obtains in *both* island and non-island conditions, suggesting that parsers try to posit the gap within the RC island as well, i.e., no RC island effects were found. This study provides the first empirical evidence that Vietnamese lacks an RC island constraint.

Exp.2 compares processing of English RC island structures by 3 groups: (1) an <u>English native control group</u> (n=54), (2) an <u>L1 Spanish-L2 English L2 control group</u> (n=39) (Spanish has a similar RC constraint as English), and (3) an <u>L1 Vietnamese-L2 English test group</u> (n=37) (Fig.2). Both L2 groups were living in their home country; their proficiency were tested. The results suggest that Vietnamese group scored significantly higher than the Spanish group ($W=1666$, $p<.0001$).

L1 English group: At spillover region 1, we observed a plausibility × island interaction (ß = -0.09, SE = 0.04, df = 1146, t = -2.129, p = .03), showing a slowdown only for implausible, non-island sentences (ß=0.07, SE=0.03, df=565, t=2.2,p=0.03). At the disambiguating region, a main effect of island status was also found (ß = -0.14, SE = 0.02, df = 1204, t = -6.37, p < .0001). These effects show sensitivity to the island constraint, in line with earlier AFS studies in English (e.g., [7]).

L1 Spanish group: We found a main effect of plausibility at spillover 1 (ß=-0.0.66, SE=0.28, df=853.3, t=-2.4, p=0.02); a nested analysis suggests that only the implausible conditions in non-island sentences are slowed down (ß =0.09, SE=0.03, df=386, t=2.41, p=0.02). At the disambiguating region, there is a main effect of island (ß=-0.14, SE=0.02,df=879.8, t=-6.1, p<.0001), similar to the English group. So, both L1 English speakers and L1 Spanish speakers reading English show sensitivity to the RC island constraint, as indicated by the asymmetrical plausibility effects.

L1 Vietnamese group: At spillover1, there are no significant effects, while at the disambiguating region, we found a main effect of plausibility (ß=-0.22, SE=0.03,df=795.6, t=-6.9, p<.0001): RTs are slower in both plausibility conditions, regardless of island status. **In sum**, we found evidence of island sensitivity in the English and Spanish groups, but not the Vietnamese group.

Discussion: By comparing speakers of English, Spanish, and Vietnamese, our study tests whether the absence of island constraints in a speaker's L1 shapes L2 processing of island structures. Our findings support current accounts of language transfer during processing [4], showing that even abstract structural constraints from the L1—constraints not reflected in surface word order—can influence L2 parsing. More broadly, these results contribute to models of bilingual language processing that view bilingual linguistic constraints as dynamic, interacting systems.

(1) Who did Jane see (a)…… Max with (b)_____?
(2) *What did Mary meet [the man who bought ___]?
(3) Cô gái [RC mà An không biết [ai đã bắt cóc___]] vừa trở về.
The girl [RC that An does not know [who PST kidnap___]] just return.
'The girl that An doesn't know who kidnapped has just returned.'

Condition	Sentence					
Non-island/ plausible	Quyển sách mà The book that	ông tác giả the author	**viết** **write**	một cách nhiệt tình passionately	về about	… …
Non-island/ implausible	Thành phố mà The city that	ông tác giả the author	**viết** **write**	một cách nhiệt tình passionately	về about	… …
Island/ plausible	Quyển sách mà The book that	ông tác giả mà the author who	**viết** **write**	một cách nhiệt tình passionately	nhắc tới… mention…	
Island/ implausible	Thành phố mà The city that	ông tác giả mà the author who	**viết** **write**	một cách nhiệt tình passionately	nhắc tới… mention…	
..được đề cử giải thưởng. ...PASS nominate prize.						

Table 1. *Exp.1 Vietnamese (**critical region**: write, spillover: passionately, disambig:about/mention)*

Non-island/plausible	The book that the author **wrote** regularly about was named for an explorer.
Non-island/ implausible	The city that the author **wrote** regularly about was named for an explorer.
Island/ plausible	The book that the author who **wrote** regularly saw was named for an explorer.
Island/implausible	The city that the author who **wrote** regularly saw was named for an explorer.

Table 2. *Exp.2 English (**critical region**: wrote, spillover1: regularly, disambig: about/saw)*

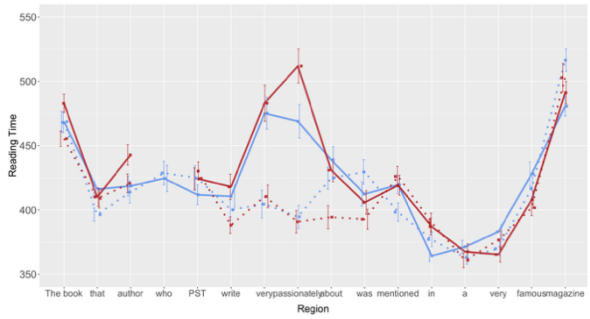
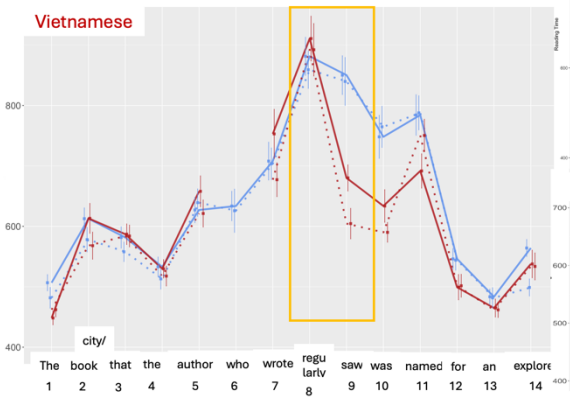


Fig.1. Raw RTs. Native Vietnamese speakers reading Vietnamese



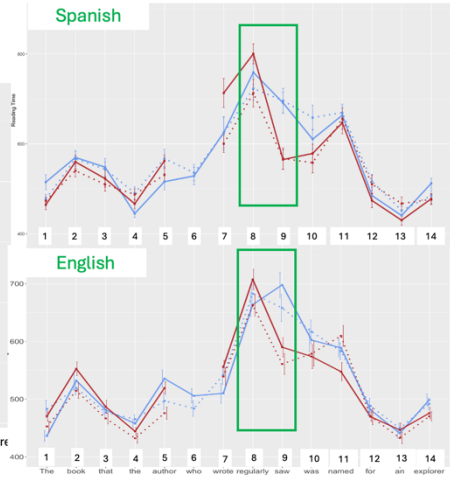


Fig.2. Raw RTs. English/Spanish/ Vietnamese groups reading English

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